

## REVIEW EXERCISE 05

**Question 1.** Given a knowledge base KB as follows,  $\{P \rightarrow R, \neg S \rightarrow P, \neg S, R \rightarrow Q\}$ . Consider the pseudo-code function PL-RESOLUTION given in the lecture to check whether KB entails Q.

Present your work to the table below, in which the first column contains  $KB \wedge \neg \alpha$  in CNF, and every of the next columns includes new sentences added to KB after each loop. Note that

- Duplicated sentences are omitted from the table
- Circle the unit clauses that lead to the contradiction and hence the function ends successfully, if possible
- Process the clauses in order, that is first pair clause 1 with clause 2, 3, 4... then pair clause 2 with clause 3, 4,... and so on.

| CNF sentences      | Loop 1             | Loop 2         | Loop 3 | Loop 4 |
|--------------------|--------------------|----------------|--------|--------|
| 1) $\neg P \vee R$ | 6) $R \vee S$      | 10) $R$        |        |        |
| 2) $P \vee S$      | 7) $\neg P \vee Q$ | 11) $\neg P$   |        |        |
| 3) $\neg S$        | 8) $P$             | 12) $S \vee Q$ |        |        |
| 4) $Q \vee \neg R$ | 9) $\neg R$        | 13) $S$        |        |        |
| 5) $\neg Q$        |                    | 14) $Q$        |        |        |

Circle the correct option, IS or IS NOT.

Following the result of resolution, the sentence Q **IS** / IS NOT entailed by KB.

**Question 2.** Repeat Question 1. but this time you check whether KB entails  $\neg Q$ .

| CNF sentences      | Loop 1             | Loop 2         | Loop 3 | Loop 4 |
|--------------------|--------------------|----------------|--------|--------|
| 1) $\neg P \vee R$ | 6) $R \vee S$      | 9) $R$         |        |        |
| 2) $P \vee S$      | 7) $\neg P \vee Q$ | 10) $S \vee Q$ |        |        |
| 3) $\neg S$        | 8) $P$             | 11) $Q$        |        |        |
| 4) $Q \vee \neg R$ |                    |                |        |        |
| 5) $Q$             |                    |                |        |        |

There is no new clause generated after Loop 2. There is no contradiction, either. Thus, KB does not entails  $\neg Q$ .

**Question 3.** Are the above problems solved by using Forward chaining or Backward chaining? Give your reason.

~~NO, there exists a sentence that is not a definite clause, i.e.  $\neg S \Rightarrow P$~~

A

**Question 4.** Consider the following text. "*Heather attended the meeting* or Heather was not invited. If *the boss wanted Heather at the meeting*, then *she was invited*. Heather did not attend the meeting. If the boss did not want Heather there, and the boss did not invite her there, then *she is going to be fired*."

Use resolution to prove that **Heather is going to be fired**. *Hint: clauses in italic are good candidates for propositions.*

Let each of following propositions denote the facts represented in the corresponding clause.

- Proposition A represents for "Heather attended the meeting."
- Proposition I represents for "Heather was invited."
- Proposition W represents for "The boss wanted Heather at the meeting."
- Proposition F represents for "Heather is going to be fired."

Then the propositional KB in CNF will be

- 1)  $A \vee \neg I$
- 2)  $\neg W \vee I$
- 3)  $\neg A$
- 4)  $W \vee I \vee F$

Apply resolution to  $KB \wedge \neg\alpha$

- 5)  $\neg F$  ..... Negation of conclusion
- 6)  $W \vee I$  ..... from sentences 4 and 5
- 7)  $I$  ..... from sentences 2 and 6
- 8)  $A$  ..... from sentences 1 and 7
- 9)  $\bullet$  ..... from sentences 3 and 8

Conclusion: Therefore, Heather is going to be fired.

**Question 5.** Consider the following knowledge base of definite clauses.

- |                                |                      |
|--------------------------------|----------------------|
| 1. $C \wedge D \rightarrow Y$  | 5. $B$               |
| 2. $R \wedge Z \rightarrow C$  | 6. $R \rightarrow D$ |
| 3. $\neg B \vee D$             | 7. $D \rightarrow R$ |
| 4. $\neg D \vee \neg R \vee Z$ |                      |

Prove **Y** using backward chaining

**Y** requires two sub-goals, C and D (from 1)

D requires the sub-goal B (from 2). B is given (from 5). Thus **D is satisfied**.

C requires two sub-goals, R and Z (from 2). R requires D (from 6), which is proved. Z requires two sub-goals, D and R, which are both proved. Thus, **C is satisfied**.

Finally, Y can be proved because all required propositions can be obtained from KB.

Prove **Y** using forward chaining (only trigger a rule once for simplicity).

8. D is obtained from 3 and 5.

9. R is obtained from 7 and 8.

10. Z is obtained from 4 and 8-9 (rewrite 4 in implication form)

10'. D is obtained **again** from 6 and 9.

11. C is obtained from 2 and 9-10

12. Y is obtained from 1 and 5-11.

Finally, Y can be proved because it is generated from KB.